

SENTINEL / ARCHITECTURE RESEARCH

Vehicle and Sensor Integration

MVP protocol scope, data model, emissions risk, and combat case studies

Prepared for the Sentinel student development team

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Executive brief

DECISION Build Sentinel around a canonical asset-and-observation model at the edge gateway. Integrate the nearest practical platform interface, simulate that interface as a separate service, and treat degraded connectivity as a normal operating state.

Connecting battlefield vehicles, ambulances, drones, and fixed sensors to a C2 platform is feasible and already proven. What is not realistic is a complete, continuously current map of every asset. Combat evidence repeatedly shows partial deployment, mixed transports, stale updates, bandwidth limits, cyber disruption, and electronic-warfare pressure.

For the MVP, Sentinel should transmit compact tracks, asset state, health, link condition, and mission events. Raw vehicle buses and high-rate payload data stay local unless a named integration requires them. The product value is a defensible, time-aware picture of what is known, how fresh it is, and how confident the system should be.

Recommended MVP cut

- Keep HTTP/JSON and WebSocket as Sentinel's internal application contract.
- Use MAVLink 2 as the first real UAV adapter and NDJSON as the deterministic replay/simulation fixture.
- Add SAPIENT/Protobuf only for a direct counter-UAS sensor integration; add ONVIF Profile T plus RTSP only for a direct EO/IR camera integration.
- Model connectivity explicitly: connected, degraded, intermittent, disconnected, emissions-restricted, and suspected-compromise.
- Keep LiDAR, CAN, DroneCAN, ROS 2/DDS, and vendor payload buses behind the vehicle or gateway boundary.
- Defer Link 16, ASTERIX, CoT/TAK, MQTT, Modbus, NMEA, LoRaWAN, and generic gRPC until a named customer, sensor, or partner creates a concrete requirement.

REALITY CHECK Encryption can protect content, but it cannot make a radio transmission invisible. Sentinel must support silence, delay, store-and-forward, alternate bearers, stale-track visualization, and local autonomy.

1. System boundary and handshake

The C2 platform should not speak every low-level sensor or vehicle protocol directly. The edge gateway is the translation and trust boundary. It collects local data, normalizes it, applies policy and buffering, and exposes a stable Sentinel contract over whichever transport is available.

Target runtime topology

VEHICLE / SENSOR SIDE

Local sensors and autopilot -> vendor or vehicle interfaces -> Sentinel edge gateway

TACTICAL NETWORK

Protected IP bearer(s), intermittent links, store-and-forward, optional silence

C2 SIDE

Sentinel ingest -> validation -> track/asset fusion -> UI and mission services

SIMULATION SIDE

Gazebo + standalone sensor simulator -> same edge gateway contract as real hardware

Handshake sequence

1. Discover and identify. The gateway recognizes a device or simulated adapter, records manufacturer/model/firmware, and assigns a stable asset or sensor identity.
2. Authenticate and authorize. The device or upstream gateway proves identity; Sentinel applies allowlists, role and command permissions, and certificate or key policy.
3. Negotiate capability. The adapter reports supported observations, commands, coordinate frames, units, rates, payload schemas, and health semantics.
4. Synchronize time and position. The gateway records clock quality, time source, coordinate reference, and uncertainty before accepting observations.
5. Stream and acknowledge. Telemetry and observations carry sequence, observed time, received time, quality, and source provenance. Commands carry expiry and acknowledgement.
6. Degrade safely. On link loss, the gateway buffers approved events, stops expired commands, preserves local control, and reports state once communication returns.

Separation of responsibilities

Component	Owns	Does not own
Vehicle/autopilot	Stabilization, navigation, safety interlocks, local payload control	C2-wide fusion or common operating picture
Sensor simulator	Synthetic detections, noise, latency, false alarms, sensor placement	Trusted ground truth exposure to the UI
Edge gateway	Adapters, normalization, identity, buffering, policy, link health	Flight-control loops or raw-sensor visualization by default

Component	Owns	Does not own
Sentinel C2	Asset state, fusion, tasking workflow, confidence and staleness UI	Vendor-specific low-level buses
Gazebo	Physics and vehicle/sensor truth for simulation	Production integration contract

2. Asset and sensor taxonomy

Sentinel should model platforms as assets with composable components, not as a flat list of sensor icons. A UAV, ground vehicle, manned aircraft, ambulance, or fixed site shares a common operational core and then adds mission-specific profiles.

Common asset profile

Component	Minimum C2 data
Identity	Asset ID, callsign, type, affiliation, trust status, software/firmware
Kinematics	Position, altitude, velocity, heading, orientation, covariance or accuracy
Readiness	Power/fuel, endurance, faults, temperature, payload availability
Communications	Bearer, link state, signal/quality indicator, latency, last contact, route
Mission/autonomy	Mode, task, route/waypoints, command status, geofence or safety state
Data quality	Observed time, received time, age, confidence, provenance

Mission sensor profiles

Profile	C2-relevant output	MVP treatment
EO/IR	Video/image reference, detection, classification, line of sight, gimbal pose	Adapter only if camera is integrated; stream metadata separately from video
Radar	Plot/track, range, bearing, elevation, radial velocity, track quality	Simulate normalized tracks; SAPIENT when a real C-UAS sensor is named
RF/ESM	Emitter detection, frequency band, bearing, confidence, identity hypothesis	Simulate bearings/events; avoid raw spectrum in the C2 core

Mission sensor profiles (continued)

Profile	C2-relevant output	MVP treatment
Cooperative ID	Transponder or friendly identity, position, status	Normalize into asset identity and trust
Threat warning	Alert type, bearing, severity, timestamp, confidence	Represent as event plus affected asset
Acoustic/UGS	Event, bearing or zone, class hypothesis, confidence	Optional demo modality; event-level data only
Weather/ environment	Wind, visibility, precipitation, pressure, hazards	Useful for route/endurance and sensor-performance context

LIDAR LiDAR is useful for local navigation, obstacle avoidance, terrain mapping, landing, and perception. It is usually not a top-level C2 sensor feed. Sentinel should ingest derived objects, hazards, maps, or clearance state, not the point cloud by default.

Ambulance and emergency-vehicle profile

An ambulance is a legitimate C2 asset, but the operational picture and clinical record must be separated. C2 needs location, availability, route, capacity, protection status, link state, and high-level casualty-load indicators. Patient identity, clinical notes, and detailed physiological data belong in a restricted medical system and should cross the C2 boundary only through deliberate policy.

- Operational channel: position, movement, mission status, destination, seats/stretchers capacity, fuel/power, faults, and last contact.
- Medical summary channel: aggregate patient count, triage categories, requested capability, contamination or isolation flags, and time sensitivity.
- Restricted clinical channel: identifiable patient data and detailed observations using healthcare schemas such as FHIR or a jurisdiction-specific EMS standard, with stricter access and audit.

3. Canonical data shape

Every adapter should translate into a small set of stable envelopes. This keeps the UI, fusion service, and simulator independent of MAVLink, ONVIF, SAPIENT, vendor APIs, or future partner platforms.

Core envelopes

Envelope	Required fields	Purpose
AssetState	assetId, observedAt, receivedAt, pose, motion, mode, readiness, linkState, source	Friendly/platform state and its quality
Observation	observationId, sensorId, observedAt, geometry, measurement, classification, confidence, source	Unassociated sensor evidence

Envelope	Required fields	Purpose
Track	trackId, state vector, covariance, classification, affiliation, confidence, contributingSources	Fused operational object
Event	eventId, type, severity, location/asset, observedAt, confidence, source	Threat, fault, medical, mission, or policy alert
Command	commandId, targetId, issuedAt, expiresAt, parameters, authority, acknowledgement	Safe, auditable tasking
AdapterHealth	adapterId, device, firmware, heartbeat, faults, queueDepth, clockQuality	Integration and gateway observability

Freshness and uncertainty are first-class

- Store observedAt and receivedAt separately. Arrival time is not measurement time.
- Expose ageSeconds, source, confidence, coordinate uncertainty, and link state to the UI.
- Render stale assets as ghosts or uncertainty regions instead of silently freezing a green icon.
- Increase uncertainty with time according to the asset's last known speed, manoeuvrability, and source quality.
- Reject or flag impossible time jumps, coordinate discontinuities, duplicate sequence numbers, and identity conflicts.

RECOMMENDED STATES CONNECTED, DEGRADED, INTERMITTENT, DISCONNECTED, EMISSIONS_RESTRICTED, and SUSPECTED_COMPROMISE. These states should drive UI treatment, command eligibility, and buffering policy.

4. Protocol scope and adapter strategy

The MVP should minimize integration surface area. Protocol support is justified by a concrete adjacent system, not by theoretical completeness.

Protocol scope matrix

Priority	Protocol/interface	Use in Sentinel
Must	HTTP/JSON + WebSocket	UI/control APIs and live normalized event delivery inside the current stack
Must	MAVLink 2	First UAV/autopilot integration; telemetry, mission state, command acknowledgement
Must	NDJSON replay	Deterministic simulation fixtures, test capture, fault injection, demos
Conditional	SAPIENT/Protobuf	Direct autonomous counter-UAS sensor or effector integration
Conditional	ONVIF Profile T + RTSP	Direct EO/IR camera control and media
Conditional	Vendor/platform API	ParallaxOS, Lattice, Picogrid, Skynode, or named sensor gateway
Conditional	CoT/TAK	Only when TAK interoperability is an explicit demo or customer requirement
Conditional	ASTERIX	Only for a named surveillance radar or air-traffic feed
Defer	Link 16	Do not implement; use an accredited gateway if a future program requires it
Defer	MQTT, Modbus, NMEA, LoRaWAN	Add as isolated adapters only when selected hardware requires them
Local only	CAN/J1939, DroneCAN, ROS 2/DDS	Keep behind the vehicle/gateway boundary; export normalized state

Adapter contract

- Identity: device, adapter, asset, sensor, manufacturer, model, firmware, certificate/key reference.
- Capabilities: observation and command types, units, frames, rates, limits, media endpoints, safety constraints.
- Lifecycle: discover, connect, authenticate, configure, start, health, stop, recover.

- Normalization: source message plus canonical envelope, with traceable conversion and no hidden unit changes.
- Resilience: bounded queue, back-pressure, retry policy, deduplication, sequence tracking, store-and-forward.
- Testability: captured fixtures, simulated faults, schema validation, time-skew and packet-loss cases.

5. Real-platform benchmark

The market confirms the architectural boundary: mature systems expose entities, tasks, tracks, video, and normalized sensor observations, while aircraft and vehicle systems retain their own real-time control and payload buses.

Platform	Publicly evidenced role	Implication for Sentinel
Anduril Lattice	Common entity model and task-oriented integration across sensors/actuators	Closest overlap at the C2 data-model and orchestration layer
Picogrid	Edge hardware/software nodes connecting heterogeneous sensors and actuators	Closest match to the gateway-plus-C2 architecture
Auterion Skynode	Onboard mission computer and autopilot ecosystem with MAVLink-facing integration	Complementary UAV layer; connect through MAVLink/API, do not replace it
ParallaxOS	Vendor-positioned heterogeneous sensor fusion and effector/tasking layer	Direct overlap; avoid duplicating protocol breadth without a target integration
SAPIENT ecosystem	Open autonomous sensor/effector messaging for counter-UAS	Best optional standard for a real C-UAS demo
TAK/CoT	Shared situational awareness and tactical data exchange	Useful interoperability edge, not the canonical internal model

Representative sensor suites

- MQ-9 Reaper public descriptions combine multi-spectral EO/IR, laser designation/ranging, and synthetic-aperture radar. Sentinel should receive selected platform state, payload state, tracks, imagery references, and mission events rather than raw internal buses [1].
- F-35 mission systems demonstrate tightly integrated radar, electro-optical, electronic-warfare, navigation, and communications functions. This reinforces the need to treat the aircraft as a fused platform source, not as a bag of independent C2 sensors [2].
- US Army VICTORY shows how a vehicle architecture can share position, time, and data services while reducing duplicated hardware. Sentinel should sit above that type of vehicle integration layer [3].

Where Sentinel should differentiate

Sentinel does not need to claim broader protocol support than established vendors. A credible student MVP can differentiate through transparent uncertainty, low-cost simulation, adapter conformance tests, cost-asymmetry analytics, and a deliberate degraded-network experience. The best integration strategy is to connect to an existing gateway or mission computer when one exists, and write a narrow adapter only where it does not.

6. Battlefield feasibility and emissions risk

Vehicle connectivity is operationally valuable because it reduces position-reporting burden, supports navigation, improves coordination, and gives commanders a shared picture. It also creates detectable and attackable communications dependencies. The correct conclusion is not 'never connect'; it is 'connect selectively, with disciplined emissions and graceful degradation.'

Risk matrix

Risk	What it means	MVP design response
Detection/geolocation	An adversary can detect or locate emissions even if payloads are encrypted	Silence mode, burst transmission, configurable rates, visible emissions state
Jamming	Bearer becomes slow, intermittent, or unavailable	Multiple bearers, queues, back-pressure, stale state, local autonomy
Spoofing/identity abuse	False or captured source injects believable data	Mutual authentication, allowlists, provenance, anomaly flags, revocation
Cyber compromise	Gateway, account, or service becomes untrusted	Least privilege, signed updates, segmentation, audit, suspected-compromise state
Bandwidth exhaustion	Video or excessive telemetry crowds out essential state	Priority classes, metadata-first video, rate control, bounded payloads
Partial deployment	Only some assets have compatible kits or connectivity	Explicit coverage gaps and human/manual reports

SECURITY PRINCIPLE A secure radio is not a stealth radio. Sentinel's UI should never imply that encrypted connectivity is free of tactical exposure.

Operational posture for the MVP

- Send essential state and events at the lowest useful rate; do not default to continuous high-rate streaming.
- Make emissions restrictions a commander/operator policy input, not an invisible network failure.

- Allow assets to remain operational when disconnected; queue only information that remains useful after delay.
- Expire commands and require acknowledgement. Never execute a stale task merely because a link recovered.
- Show source, time, confidence, and communication state at every decision point.

7. Combat case studies

Iraq: Blue Force Tracking was useful, incomplete, and bandwidth-bound

During Operation Iraqi Freedom, FBCB2/Blue Force Tracking gave equipped units a shared friendly-force picture, supported movement through dust and poor visibility, and reduced some voice position reporting. Public after-action material also records limited fielding, mixed systems, bandwidth constraints, and update intervals that could leave a partial or stale picture. One contemporary Armor article described 1,189 BFT packages across participating units and reported early update behavior around five minutes or 800 metres [12].

The 3/2 Stryker Brigade Combat Team's Iraq experience further shows that digital tracking did not replace communications diversity. Its long movement relied on BFT alongside tactical satellite, HF, and other bearers [13]. A GAO review found multiple incompatible blue-force tracking systems during OIF, forcing improvised bridges and demonstrating that interoperability is an operational problem, not a checkbox [14].

SENTINEL LESSON Design for mixed fleets and an incomplete picture. A stale or missing asset is a first-class state, and a gateway bridge is a normal deployment component.

Afghanistan: incremental upgrades and mission-specific digital coordination

The US Army's Joint Capabilities Release/JBC-P upgrade in Afghanistan illustrates practical fleet rollout rather than universal adoption: an Army report described upgrades for 275 vehicles and 32 tactical operations centre kits by February 2013 [15]. Years later, a TAK-based portal reportedly supported operators during the 2021 Afghanistan evacuation, showing the value of rapidly assembled, mission-specific situational-awareness tools [16].

SENTINEL LESSON Ship adapters and kits in useful increments. Support manual/operator-sourced events and temporary partner interfaces instead of assuming every platform arrives digitally integrated.

Ukraine: software-defined C2 at scale, under electronic attack

Ukraine's DELTA system is publicly described as a combat-proven situational-awareness and mission-management environment used across defence-force levels. Ukrainian and NATO sources describe integration of sensors, trackers, radars, satellites, UAV and fixed-camera streams, and robotic systems, with access from ordinary computing devices [17][18]. A January 2025 announcement said DELTA Mission Control was being used by more than 300 units for roughly 900 missions per day and had supported more than 200,000 missions [19].

The same war demonstrates the fragility of communications. RUSI reporting documents heavy dependence on commercial satellite connectivity, jamming and cyber attacks, and the danger of radio-frequency geolocation [20][21][22]. Tactical systems such as Kropyva and GIS Arta are also cited in public analysis as part of a distributed digital fires and coordination ecosystem [23].

SENTINEL LESSON The realistic product is a distributed, multi-source picture that survives disruption, not a permanently synchronized cloud dashboard. Commercial and tactical bearers can be powerful, but neither availability nor concealment is guaranteed.

Cross-case synthesis

Observed pattern	Evidence across cases	Sentinel requirement
Partial deployment	Different units and vehicles receive different kits and versions	Unknown/unintegrated assets and manual reports
Mixed bearers	Satellite, tactical radio, cellular/commercial, and local networks coexist	Transport-agnostic gateway and route visibility
Stale data	Update intervals and outages make the picture lag reality	Age, uncertainty growth, ghosting
Interoperability gaps	Multiple systems require gateways or improvised bridges	Canonical model and narrow adapters
EW/cyber pressure	Jamming, geolocation, service disruption, and compromise are credible	Silence, degradation, revocation, local autonomy
High operational value	Navigation, coordination, targeting workflow, and reduced voice burden	Preserve essential state and tasking under constraint

8. MVP demonstration and acceptance criteria

The demonstration should prove that Sentinel behaves honestly when the network and sources are imperfect. The sensor simulator must be a separate process connected through the same edge gateway contract intended for real sensors and drones.

Three-scene demo

1. Connected baseline. Gazebo UAV, simulated radar/RF/EO events, and an ambulance/ground vehicle publish through the edge gateway. The UI shows source, freshness, confidence, readiness, and link state.

2. Jamming and partition. Latency and packet loss rise; one asset enters emissions-restricted mode; another disconnects. Sentinel grows uncertainty, expires a command, buffers useful events, and avoids showing frozen certainty.

3. Recovery and trust fault. The link returns and the queue drains with original observation times. A conflicting identity or impossible jump triggers suspected-compromise treatment and prevents automatic tasking.

Acceptance criteria

Gate	Pass condition
Contract parity	Real, replay, Gazebo, and synthetic adapters produce the same validated canonical envelopes.
Truth isolation	The UI never reads simulator ground truth directly.
Time semantics	Observed and received times remain distinct through buffering and replay.
Degradation	Loss, delay, reordering, duplication, time skew, and recovery are visible and testable.
Command safety	Commands have authority, expiry, acknowledgement, and disconnected behavior.
Data minimization	Raw point clouds, vehicle buses, and unrequested clinical data do not enter the core C2 stream.
Protocol restraint	Only the three must-have interfaces are implemented without a named conditional integration.
Operator honesty	Every asset/track view exposes freshness, confidence, provenance, and link state.

Implementation order

1. Freeze canonical schemas and connectivity semantics.
2. Finish the standalone sensor simulator and NDJSON capture/replay path.
3. Implement the edge gateway lifecycle, buffering, validation, and health telemetry.
4. Add MAVLink 2 telemetry and command acknowledgement against a software-in-the-loop vehicle.
5. Update the UI for staleness, uncertainty, emissions restriction, and suspected compromise.
6. Add one conditional real-sensor adapter only after selecting the actual demo hardware.

BOTTOM LINE Vehicle-connected C2 is not a fairytale. The fairytale is pretending connectivity is universal, continuous, and tactically free. Sentinel should make uncertainty and network reality part of the product.

Appendix A. Source register

Public sources used to benchmark platform roles, sensor suites, interoperability, and battlefield deployments. Vendor claims are treated as descriptions of stated capability, not independent performance validation.

- [1] [US Air Force - MQ-9 Reaper fact sheet](#). Public payload and mission-system overview.
- [2] [Lockheed Martin - F-35 Mission Systems design and verification](#). Integrated mission-system architecture and sensor functions.
- [3] [US Army - VICTORY open systems architecture for fighting vehicles](#). Vehicle integration and shared data/service approach.
- [4] [Auterion - Skynode peripheral integration](#). Onboard mission-computer and peripheral interfaces.
- [5] [MAVLink - Common message set](#). Public UAV telemetry and command vocabulary.
- [6] [Anduril - Lattice entities overview](#). Public entity-model integration guidance.
- [7] [Picogrid - Orion API getting started](#). Public API entry point for the edge/C2 platform.
- [8] [Picogrid - Echodyne partner integration](#). Example radar ecosystem integration.
- [9] [UK Government - SAPIENT autonomous sensor system](#). Open autonomous sensor/effector integration standard.
- [10] [ONVIF - Profile T specification](#). Standardized IP video streaming and control profile.
- [11] [MITRE - Cursor on Target overview](#). Tactical event exchange background.
- [12] [US Army Armor - Blue Force Tracking in Operation Iraqi Freedom](#). Contemporary field observations, scale, benefits, and constraints.
- [13] [RAND - 3/2 Stryker Brigade Combat Team in Iraq](#). Operational movement and communications mix.
- [14] [US GAO - Battlefield automation interoperability in Operation Iraqi Freedom](#). Incompatible tracking systems and interoperability findings.
- [15] [US Army - Blue Force Tracking upgrade in Afghanistan](#). Vehicle and TOC kit rollout figures.
- [16] [US Air Force Research Laboratory - TAK support during Afghanistan evacuation](#). Mission-specific digital coordination case.
- [17] [Ukraine Ministry of Defence - DELTA deployed across defence-force levels](#). Official description of deployment and combat use.
- [18] [NATO ACT - DELTA at CWIX](#). Interoperability and multi-source integration description.
- [19] [Ukraine Ministry of Defence - DELTA Mission Control adoption](#). Published usage and mission figures.
- [20] [RUSI - Preliminary lessons from Russia's war in Ukraine](#). Communications, commercial satellite dependence, and operational lessons.
- [21] [RUSI - Jamming and cyber attacks against space services in Ukraine](#). Jamming and cyber disruption examples.
- [22] [RUSI - Russian communications and the world of hertz](#). Radio-frequency detection and geolocation risk.
- [23] [RUSI - Tactical developments in the third year of the Russo-Ukrainian War](#). Public analysis of tactical digital systems and adaptation.
- [24] [NATO - Electronic warfare in a contested environment](#). Alliance-level electronic-warfare context.
- [25] [Ukraine Ministry of Defence - DELTA cloud and cyber architecture](#). Official discussion of DELTA development and infrastructure.